

RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD**Faculty of Engineering & Applied Sciences**

Department of Electrical & Computer Engineering

End Term Examination — Spring 2026 Semester

Course Name:	Digital Logic Design	Course Code:	EE20203 / FLB20203
Program:	BS (Software Engineering)	Section:	2
Semester:	2nd Semester	Date of Examination:	4 - June - 2026
Teacher's Name:	Syed Hassaan	Total Marks:	40
Exam Duration:	2-3 Hours	Student Name:	
SAP ID:		Obtained Marks:	

INSTRUCTIONS

- Write your Name and SAP ID on all answer sheets.
- Write page number on all answer sheets.
- All questions are compulsory. Partial credit may be awarded for incomplete but relevant work.
- Show all working/steps clearly. Final answers without supporting work may not receive full marks.
- Switch off your Mobile Phone. No electronic devices allowed.
- Neat and legible work will be given preference.

Question 1:**[12 Marks]**

Answer the following questions related to Boolean minimization using the Quine-McCluskey (QM) method:

(a) Minimize the following Boolean function using the Quine-McCluskey (tabulation) method. Show all grouping tables, indicate all Prime Implicants (PIs), construct the Prime Implicant Table, identify Essential Prime Implicants (EPIs), and write the final minimized SOP expression. **[6 Marks]**

$$F(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$$

(b) Apply the Quine-McCluskey method to the following function which includes don't-care conditions. Identify all Prime Implicants and give the minimized SOP expression. Don't-cares may be used to simplify but need not be covered. **[4 Marks]**

$$F(W, X, Y, Z) = \sum m(2, 6, 8, 9, 10) + \sum d(0, 11, 15)$$

(c) A function of 5 variables $f(A, B, C, D, E) = \sum m(1, 4, 6, 10, 20, 22, 24, 26)$ is to be minimized. Briefly explain why the Quine-McCluskey method is preferred over Karnaugh maps for this function, and list the first-column groupings (by number of 1-bits) without performing full minimization. **[2 Marks]**

Question 2:**[14 Marks]**

Answer the following questions on Combinational MSI Circuits — Adders, Subtractors, and Comparators:

(a) Design a Full Adder circuit from scratch. Derive its truth table for inputs X, Y, and Cin. Write the simplified Boolean expressions for Sum (S) and Carry-out (Cout). Draw the logic circuit using only basic gates (AND, OR, XOR). **[4 Marks]**

(b) Show how a 4-bit Adder/Subtractor circuit can be built using a 4-bit ripple carry adder and XOR gates. Explain the role of the Mode select bit M: what happens when M = 0 and when M = 1. Write the operation it performs for each case. **[4 Marks]**

(c) Using two 74x283 4-bit adder ICs, design an 8-bit adder. Show how the carry-out of the lower nibble IC connects to the carry-in of the upper nibble IC. If A = 10110011 and B = 01101110, compute A + B showing the carry propagation between the two ICs. **[3 Marks]**

(d) Two 4-bit numbers A = 1011 and B = 1101 are to be compared using a 74x85 magnitude comparator. Determine the values of the three outputs (A > B, A = B, A < B). State the values you would assign to the cascading inputs for a standalone (non-cascaded) operation. **[3 Marks]**

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Question 3:

[14 Marks]

Answer the following questions on MSI Combinational Building Blocks — Decoders, Encoders, Multiplexers, and Demultiplexers:

(a) Implement the following two Boolean functions simultaneously using a single 3-to-8 decoder and OR gates only. Clearly identify which decoder output lines you are ORing for each function. **[3 Marks]**

$$F1(A, B, C) = \Sigma m(1, 3, 5, 7) \quad \text{and} \quad F2(A, B, C) = \Sigma m(0, 2, 4, 6)$$

(b) Design a 4-to-1 Multiplexer circuit using basic logic gates. Draw its circuit diagram. Then use a 4-to-1 MUX to implement the Boolean function $F(A, B, C) = \Sigma m(0, 2, 5, 7)$, where A and B are the select lines and C is available as a data variable.

Determine the values to be connected to data inputs I0, I1, I2, and I3. **[4 Marks]**

(c) Using an 8-to-1 multiplexer (74151A), implement the function $F(A, B, C) = \Sigma m(1, 2, 4, 6, 7)$. Connect A, B, C to select lines S2, S1, S0 respectively, and determine each data input I_k (0 or 1). Verify one entry from your truth table. **[4 Marks]**

(d) A 1-to-8 demultiplexer is required but only 1-to-4 DEMUX ICs are available. Show how you would cascade two 1-to-4 DEMUX units to produce a 1-to-8 DEMUX. Label all data inputs, select lines, and output lines clearly. **[3 Marks]**

— **END OF EXAMINATION PAPER** —

Total: 40 Marks | Duration: 2 Hours | Good Luck!